

Physics 214/224, Practice Quiz 4

Primary instructor: Yurii Maravin

WID: _____ NAME: _____

Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: “On my honor as a student I have neither given nor received unauthorized aid on this academic work.”

For two quick points, circle your studio instructor:

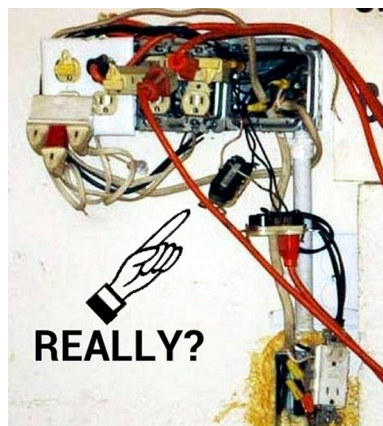
Lohman (TU 7:30) Lohman (TU 9:30) Weaver (TU 9:30)

Magrakvelidze (TU 11:30) Maravin (TU 1:30) Weaver (TU 3:30)

Magrakvelidze (WF 7:30) Magrakvelidze (WF 9:30)

Golin (WF 11:30)

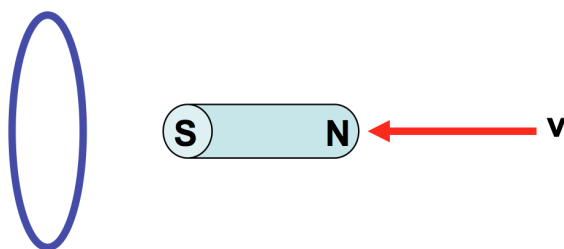
Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.

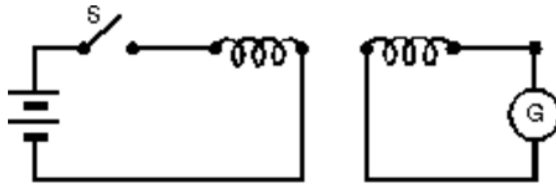


Never try this at home!

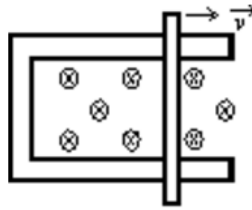
Part I: All questions to be answered on your Scantron card (48 points total)

1. (3 points) If the circular conductor undergoes thermal expansion while in a uniform magnetic field, a current is induced clockwise around it. Is the magnetic field directed into or out of the page?
 - (a) into the page
 - (b) out of the page
2. (3 points) A 2.0 T uniform magnetic field makes an angle of 30° with the z -axis. The magnetic flux through a 3.0 m^2 area of the xy plane is:
 - (a) 6.0 Wb
 - (b) 12 Wb
 - (c) 2.0 Wb
 - (d) 5.2 Wb
 - (e) 3.0 Wb
3. (3 points) Which way, if any, will current flow in the loop as the magnet is pushed in, as viewed from the right?



- (a) Clockwise
 - (b) Counter-clockwise
 - (c) No current will flow
4. (3 points) In the experiment shown in picture below:
 - (a) the current in the battery goes through G.
 - (b) since the two loops are not connected, the current in G is always zero
 - (c) there is a steady reading in G as long as S is closed
 - (d) there is a current in G just after S is opened or closed

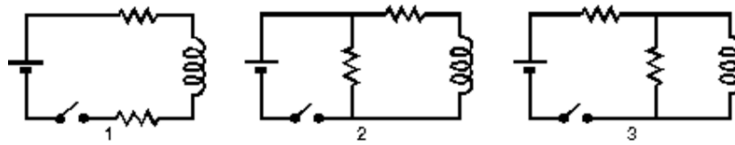
5. (3 points) A rod lies across frictionless rails in a uniform magnetic field B , as shown. The rod moves to the right with speed v . In order for the EMF around the circuit to be zero, the magnitude of the magnetic field should:



- (a) increase linearly with time
 - (b) decrease quadratically with time
 - (c) increase quadratically with time
 - (d) decrease linearly with time
 - (e) not change
6. (3 points) A long straight wire is in the plane of a rectangular conducting loop. The straight wire initially carries a constant current i in the direction shown. While the current i is being shut off, the current in the loop is:



- (a) clockwise
 - (b) counter-clockwise
 - (c) zero
 - (d) clockwise in the left side of the loop, and counter-clockwise in the right side
 - (e) counter-clockwise in the left side of the loop, and clockwise in the right side
7. (3 points) An inductance L , resistance R , and ideal battery of emf $\mathcal{E}$ are wired in series and the circuit is closed at time $t = 0$, at which time the current is zero. At any later time t the potential difference across the resistor is given by:
- (a) $\mathcal{E}(1 - e^{-Lt/R})$
 - (b) $\mathcal{E}(1 - e^{-Rt/L})$
 - (c) $\mathcal{E}(1 + e^{-Rt/L})$
 - (d) $\mathcal{E}e^{-Lt/R}$
 - (e) $\mathcal{E}e^{-Rt/L}$
8. (3 points) The diagrams show three circuits with identical batteries, identical inductors, and identical resistors. Rank them according to the current through the battery just after the switch is closed, from least to greatest.



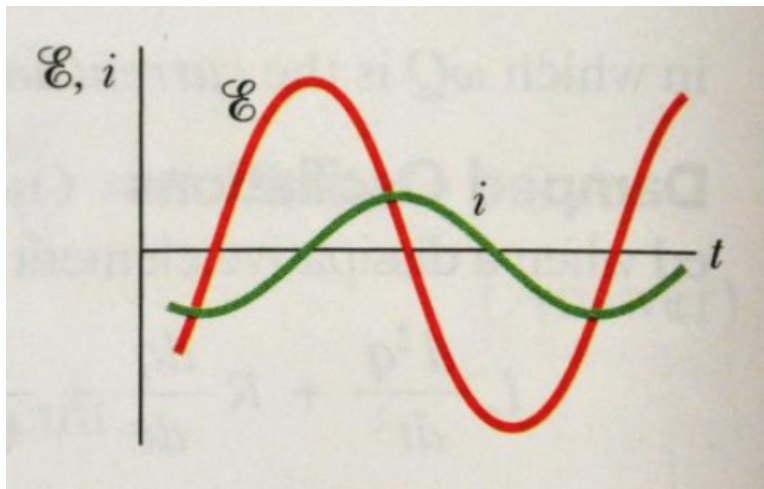
- (a) 3, 2, 1
 (b) 2, 3, 1
 (c) 1, 2, 3
 (d) 2 and 3 tie, then 1
 (e) 1, 3, 2
9. (3 points) A 5 mH inductor and a 10 mH inductor are connected in parallel. The equivalent inductance is:
- (a) 15 mH
 (b) 3.3 mH
 (c) 5 mH
 (d) 2 mH
10. (3 points) A flat coil of wire, having 5 turns, has an inductance L . The inductance of a similar coil having 20 turns is:
- (a) $4L$
 (b) L
 (c) $L/16$
 (d) $16L$
11. (3 points) An LC circuit consists of a $1 \mu\text{F}$ capacitor and a 4 mH inductor. Its oscillation frequency is approximately:
- (a) 60 Hz
 (b) 16,000 Hz
 (c) 2500 Hz
 (d) 0.025 Hz
 (e) 25 Hz
12. (3 points) A sinusoidal voltage $V(t)$ has an rms value of 100 V. Its maximum value is:
- (a) 100 V
 (b) 141 V
 (c) 70.7 V
 (d) 707 V
 (e) 200 V
13. (3 points) A 45-mH inductor is connected to an AC source of EMF with a frequency of 400 Hz and a maximum EMF of 20 V. The maximum current is:

- (a) 2300 A
- (b) 0.18 A
- (c) 1.1 A
- (d) 0 A
- (e) 360 A

14. (3 points) The impedance of an RLC series circuit is definitely increased if:

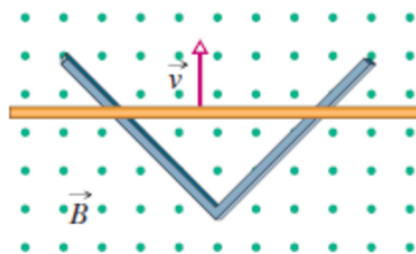
- (a) L decreases
- (b) L increases
- (c) R decreases
- (d) C decreases
- (e) R increases

15. (3 points) For the figure below, that shows the current i and a driving EMF $\mathcal{E}$ for a series RLC circuit, does the current lead or lag the EMF?



- (a) The current leads the EMF
 - (b) The current lags the EMF
 - (c) Both current and EMF are in phase
16. (3 points) For the figure above, is the circuit's load mainly capacitive or mainly inductive?
- (a) The load is mainly capacitive
 - (b) The load is mainly inductive
 - (c) None of the above

Part II: Work out this part in the space provided. Show your work!
(50 points total)



17. (10 points) In the figure above, two straight conducting rails form a right angle. A conducting bar in contact with the rails starts at the vertex at time $t = 0$ and moves with a constant velocity of 4.50 m/s along them. A magnetic field with $B = 0.350$ T is directed out of the page.

- (a) (4 points) Calculate the flux through the triangle formed by the bars at $t = 3.10$ s.

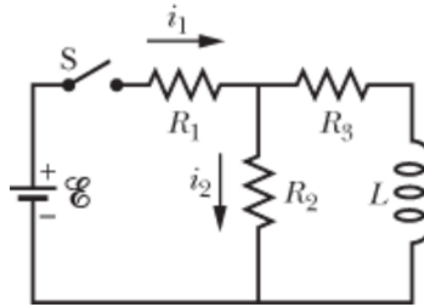
At a time t , the bar will be at $v \cdot t = 4.50 \text{ m/s} \cdot 3.10 \text{ s} = 13.95 \text{ m}$ away from the vertex. Therefore, the length of the rails will be $v \cdot t \cdot \sqrt{2} = 13.95 \cdot \sqrt{2} = 19.7 \text{ m}$, and the triangle area is $(v \cdot t \cdot \sqrt{2})^2 / 2 = (13.95 \text{ m})^2 = 195 \text{ m}^2$
The flux will be $195 \text{ m}^2 \cdot 0.35 \text{ T} = 68.1 \text{ Wb}$

- (b) (3 points) Calculate the emf around the triangle at that time.

$\text{EMF} = - \frac{d\Phi}{dt} = -B \cdot \frac{dA}{dt} = -B \cdot \frac{d[(vt)^2]}{dt} = -B \cdot 2(vt) \cdot v = -2 \cdot B \cdot v^2 \cdot t = -43.9 \text{ V}$
Or, a magnitude of 43.9 V

- (c) (3 points) If the emf $\mathcal{E} = at^n$, where a and n are constants, what is the value of n ?

EMF t , so $n = 1$



18. (12 points) In the circuit above $\mathcal{E} = 100 \text{ V}$, $R_1 = 10.0 \, \Omega$, $R_2 = 20.0 \, \Omega$, and $R_3 = 30.0 \, \Omega$, and $L = 2.00 \text{ H}$. Assume positive currents i_1 and i_2 in the directions indicated in the figure. If the current flows in the opposite direction than what is specified, indicate it as negative.

- (a) (3 points) Immediately after switch S is closed, what are currents i_1 and i_2 ?

The inductor would resist any current to flow through it, immediately after S is closed. So, the current through R_3 is zero, and $i_1 = i_2 = \text{EMF}/(R_1 + R_2)$ from Ohm's Law. Hence, $i_1 = i_2 = 100\text{V}/30 \text{ Ohm} = 3.33 \text{ A}$

- (b) (3 points) A long time after, what are currents i_1 and i_2 ?

After long time, L is essentially serve as a short, so you have a series connection of R_1 with parallel connection of R_2 and R_3 . The effective resistance of R_2 and R_3 in parallel is $20 \cdot 30 / (20 + 30) \text{ Ohms} = 12 \text{ Ohms}$, the total resistance of the circuit is $10 + 12 \text{ Ohms} = 22 \text{ Ohms}$. So the current $i_1 = 100 \text{ V} / (22 \text{ Ohms}) = 4.55 \text{ A}$, and voltage drop on R_1 is $4.55 \text{ A} \cdot 10 \text{ Ohms} = 45.5 \text{ V}$. Thus, $100 \text{ V} - 45.5 \text{ V} = 54.5 \text{ V}$ is dropped on the R_2 and R_3 . Hence, $i_2 = 54.5 \text{ V} / (20 \text{ Ohms}) = 2.73 \text{ A}$

- (c) (3 points) The switch is then reopened. Just then, what are i_1 and i_2 ?

i_1 will be zero, as the circuit is open, while i_2 will be the current that flows through the inductor at the moment of the opening the switch, i.e., $4.55 \text{ A} - 2.73 \text{ A} = 1.82 \text{ A}$. It will be exponentially decrease with the time constant of $\tau = L / (R_2 + R_3) = 0.04 \text{ sec}$

- (d) (3 points) A long time after the switch is reopened, what are i_1 and i_2 ?

As the current is exponentially decreased, after some time i_2 will become zero as well.

19. (12 points) An vertically oriented N -turn air-core solenoid with cross sectional area A and length D carries a time varying current $I = I_0 e^{-t/\tau}$. When viewed from above, the current in the solenoid flows in a clockwise direction. An M -turn circular loop of wire of radius R surrounds the solenoid and shares its axis of symmetry. Assume ideal behavior for the solenoid, and express your answers below in terms of the symbolic quantities given, plus natural constants.

- (a) (3 points) Find the absolute value of the voltage difference across the solenoid produced by its self-inductance at time $t = 0$.

The self-inductance of the solenoid is $L = \mu_0 N^2 A/D$, so from definition of

L :

$$\text{Flux} = L \cdot I = (\mu_0 N^2 A/D) \cdot I_0 e^{-t/\tau}$$

$$\text{From Faraday's law: EMF} = -d\text{Flux}/dt = -(\mu_0 N^2 A/D)/\tau \cdot I_0 e^{-t/\tau}$$

At time $t = 0$, the exponent is just equal to 1, so

$$\text{EMF} = -(\mu_0 N^2 A/D)/\tau \cdot I_0$$

The absolute value will be without a minus sign :)

- (b) (3 points) Show that the magnetic field flux from the solenoid through the circular loop at time $t = 0$ is $\Phi_B = \frac{MNA}{D} \mu_0 I_0$.

The self-inductance is $M_{12} = \mu_0 NMA/D$, so $\text{Flux}_{\text{Loop}} = M_{12} I_0 e^{-t/\tau}$

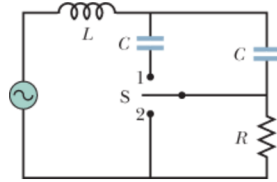
Therefore, the flux through the loop at time $t = 0$ is $\mu_0 NMA I_0/D$

- (c) (3 points) Find the absolute value of the voltage difference across the circular loop at time $t = 0$

$$|\text{EMF}_{\text{Loop}}| = |d\text{Flux}_{\text{Loop}}/dt| = d[(\mu_0 NMA/D) \cdot I_0 e^{-t/\tau}]/dt = (\mu_0 NMA/D) \cdot I_0/\tau$$

- (d) (3 points) Determine the direction that induced current will flow in the circular loop when viewed from above. Explain your answer with one or two sentences.

The current is reducing in N -turn solenoid, so the magnetic flux in both solenoid and the loop will be reducing as well. The current in the loop will flow in the same direction as in solenoid (i.e., clockwise), so that the magnetic field, generated by the loop, would oppose the decrease in the flux.



20. (16 points) In the AC circuit above, the generator supplies an EMF $V(t) = V_0 \cos(\omega_d t)$ with $V_0 = 170$ V and $\omega_d = 2\pi f_d$. The frequency $f_d = 60.0$ Hz. The value of the resistor is $R = 100$ Ω , $L = 10$ mH, and each capacitor has a capacitance of $C = 10$ μ F.

- (a) (3 points) Calculate effective capacitance of the circuit for the switch in neutral (as shown), in position 1, and position 2.

In neutral position, only one capacitance is connected, so $C_{\text{eff}} = 10$ microfarad

In position 1, two capacitors are in parallel, so $C_{\text{eff}} = 20$ microfarad ($C_{\text{par}} = C_1 + C_2$)

In position 2, only one capacitor is connected, so $C_{\text{eff}} = 10$ microfarad

- (b) (3 points) Calculate capacitive reactance of the circuit for the switch in neutral (as shown), in position 1, and position 2.

$$X_C = 1/(2\pi f C)$$

so, for neutral position and position 2: $X_C = 265$ Ohm

for position 1: $X_C = 132$ Ohm

- (c) (2 points) Calculate inductive reactance of the circuit for the switch in neutral (as shown), in position 1, and position 2.

In all three cases, the inductance is the same, and inductive reactance $X_L = 2\pi f L = 3.77$ Ohm

- (d) (2 points) Calculate the resistive reactance of the circuit for the switch in neutral (as shown), in position 1, and position 2.

Resistive reactance (resistance) in positions 1 and neutral is the same $X_R = 100$ Ohm, but in position 2, the resistor is shorted out, so $X_R = 0$ Ohm.

- (e) (3 points) What is the impedance of the circuit for the switch in neutral (as shown)?

$$Z = [(X_L - X_C)^2 + R^2]^{1/2} = 280 \text{ Ohm}$$

- (f) (3 points) What is the resonant frequency of the circuit for the switch in neutral (as shown)?

$$\omega = 1/\sqrt{LC}, \text{ so } f = 1/(2\pi \sqrt{LC}) = 503 \text{ Hz}$$